

## Training Configuration and Summary:

Dataset used: rssi\_odom\_dataset\_12apr25\_ajust.parquet

model\_type: LSTM  
group\_sec: 80  
timesteps\_orig: 60  
batch\_size: 256  
min\_units: 32  
max\_units: 256  
units\_step: 32  
dropout\_min: 0.2  
dropout\_max: 0.3  
dropout\_step: 0.1  
activation\_functions: ['tanh', 'relu', 'elu']  
optimizers: ['adam', 'rmsprop', 'Nadam']  
min\_layers: 1  
max\_layers: 2  
max\_epochs: 150  
hyperband\_iterations: 3  
group\_msec: 80  
samples\_per\_sec: 12  
forecast\_horizon\_sec: 4  
forecast\_steps: 48  
timesteps: 60

### LSTM Multi-Step Training Report

Report Generated: 2025-05-15 21:46:17

Training Start: 2025-05-13 16:39:07

Training End: 2025-05-13 16:49:16

Training Duration: 10.16 minutes

Predictions Start: 2025-05-13 16:49:16

Predictions End: 2025-05-13 16:50:59

Predictions Duration: 1.72 minutes

#### Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 1.8589

MSE: 5.5363

RMSE: 2.3529

R2 Score: 0.6674

#### Multi-Step Forecast:

Forecast Steps: 48

MAE: 0.0326

MSE: 0.0016

RMSE: 0.0401

R2 Score: 0.8262

Dataset size before down-sample: 85,269 rows  
Dataset size after down-sample: 12,829 rows

First 5 timestamps BEFORE:

1744397085760  
1744397085770  
1744397085789  
1744397085801  
1744397085817

...

Last 5 timestamps BEFORE:

1744464971366  
1744464971377  
1744464971390  
1744464971402  
1744464971414

First 5 timestamps AFTER:

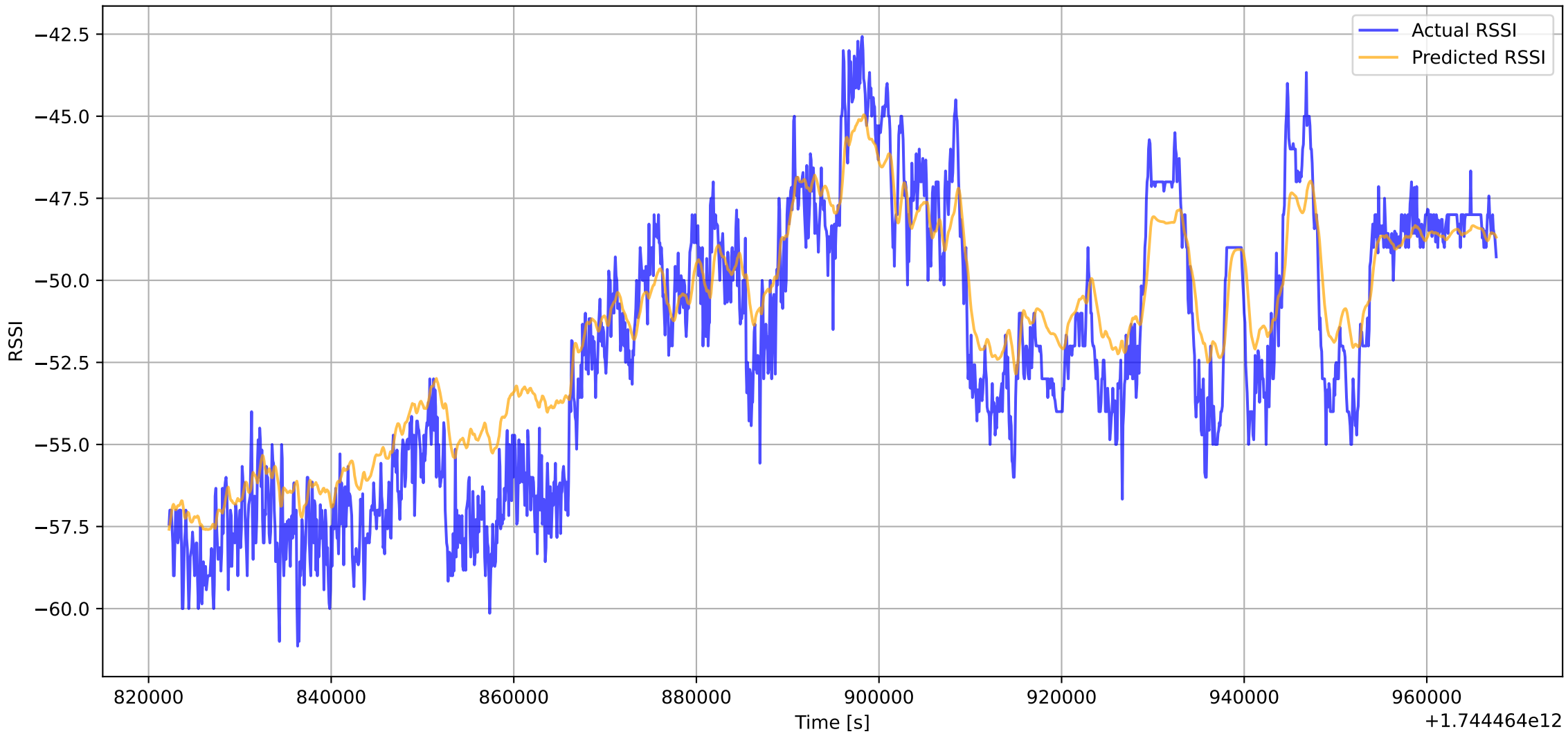
1744397085760  
1744397085841  
1744397086014  
1744397086103  
1744397086172

...

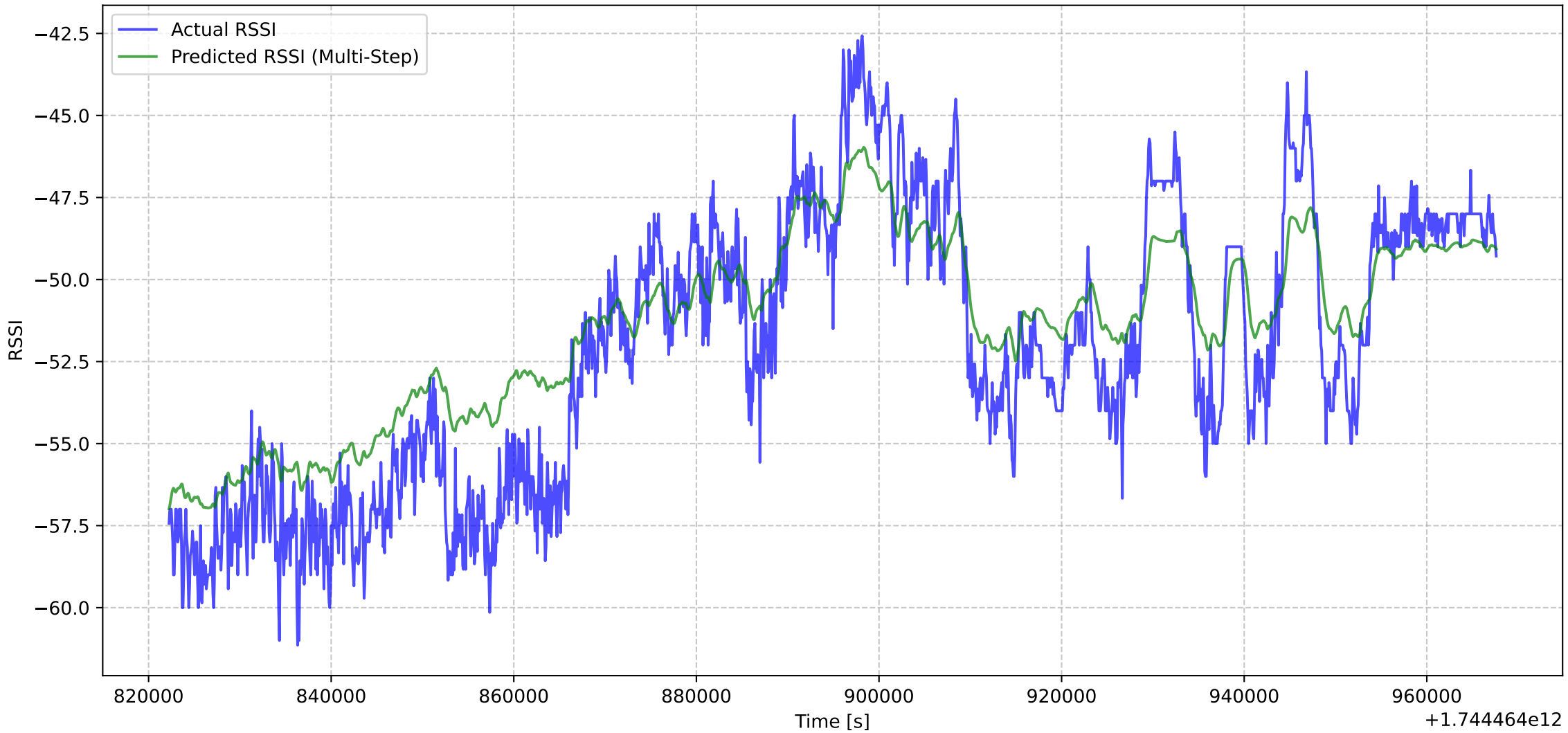
Last 5 timestamps AFTER:

1744464971041  
1744464971124  
1744464971209  
1744464971281  
1744464971366

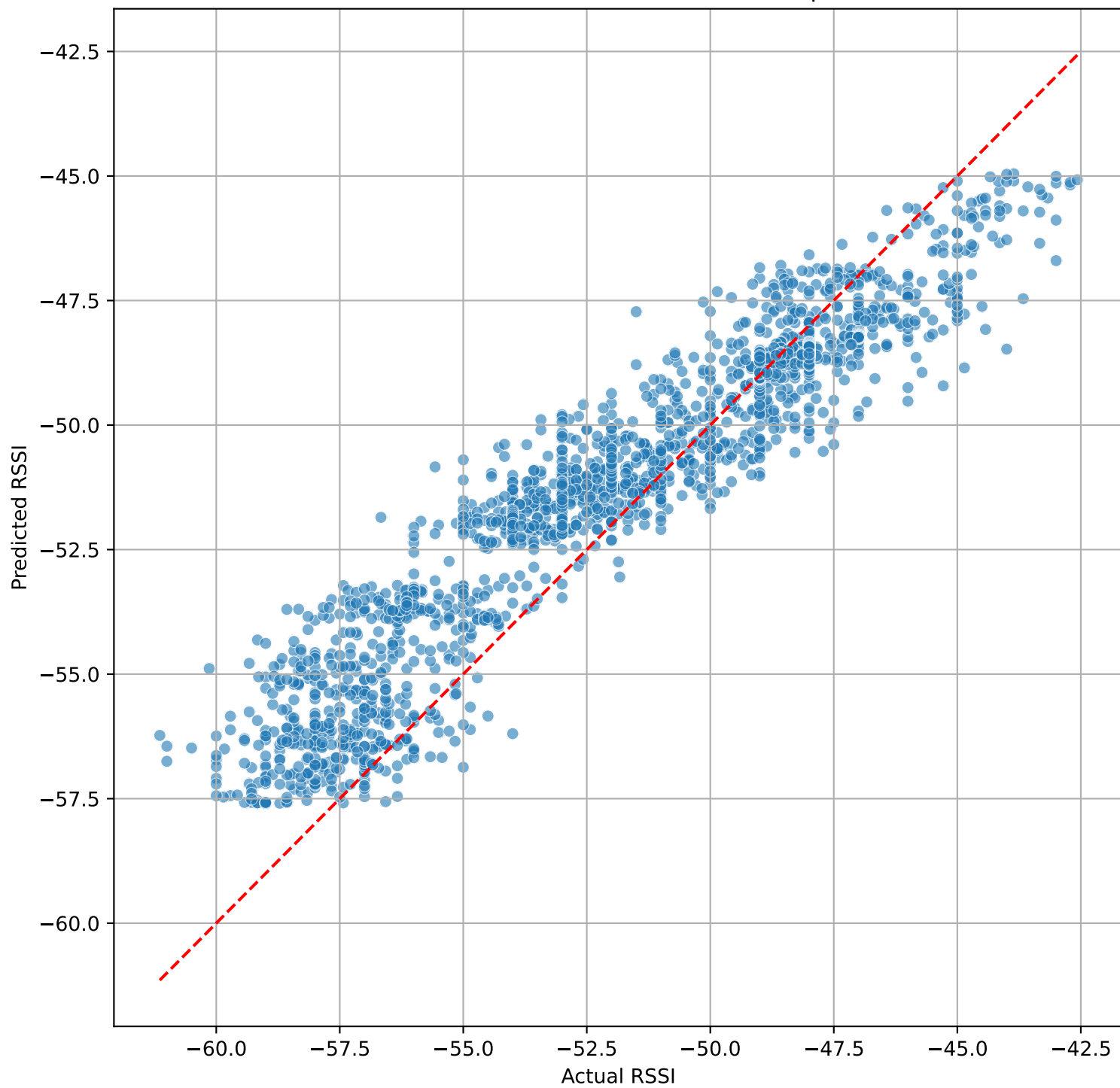
Actual vs Predicted RSSI (First Step)



Actual vs Predicted RSSI (Multi-Step (step 48 $\approx$  4.0 s))



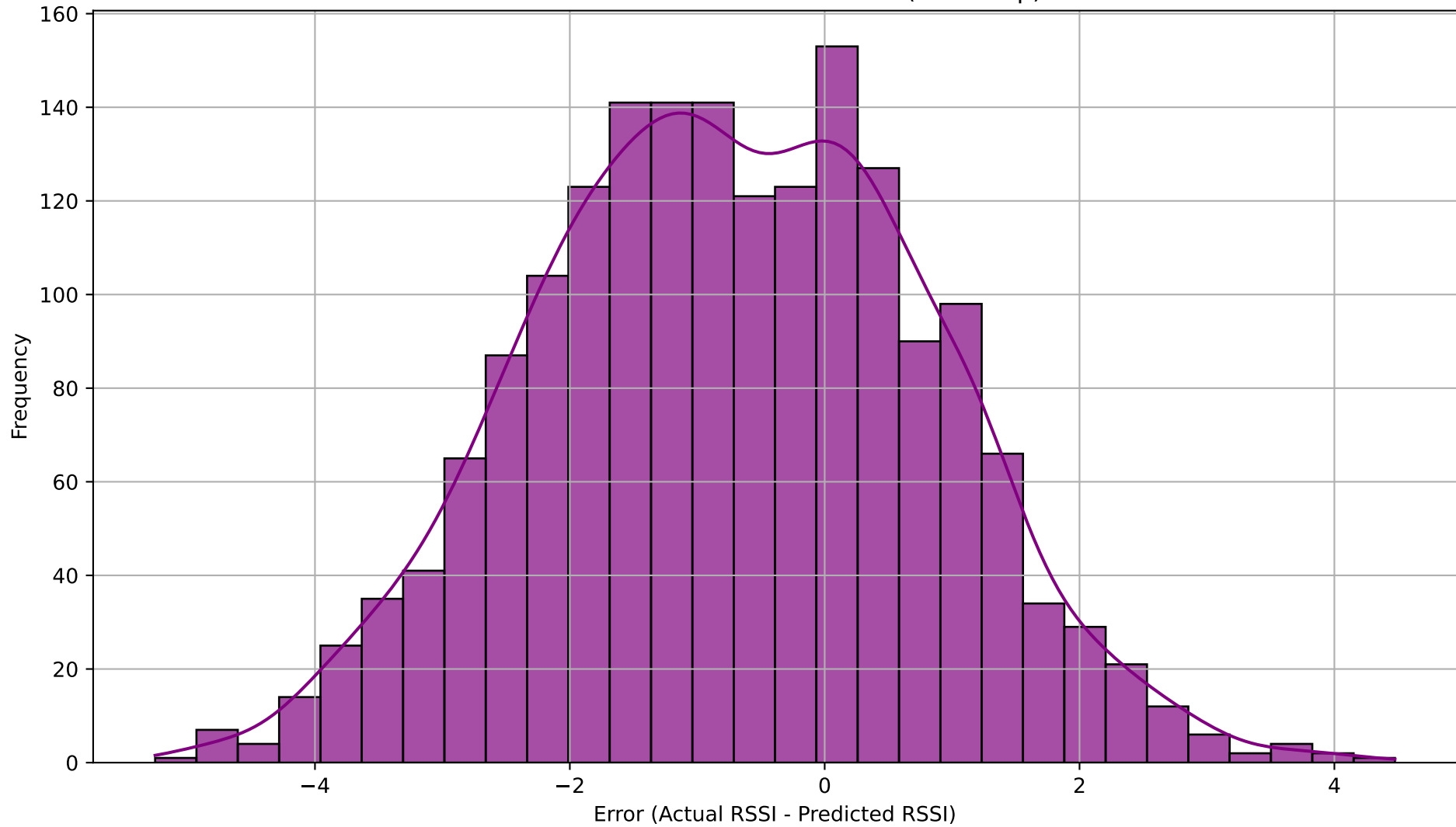
Actual vs Predicted RSSI (First Step)



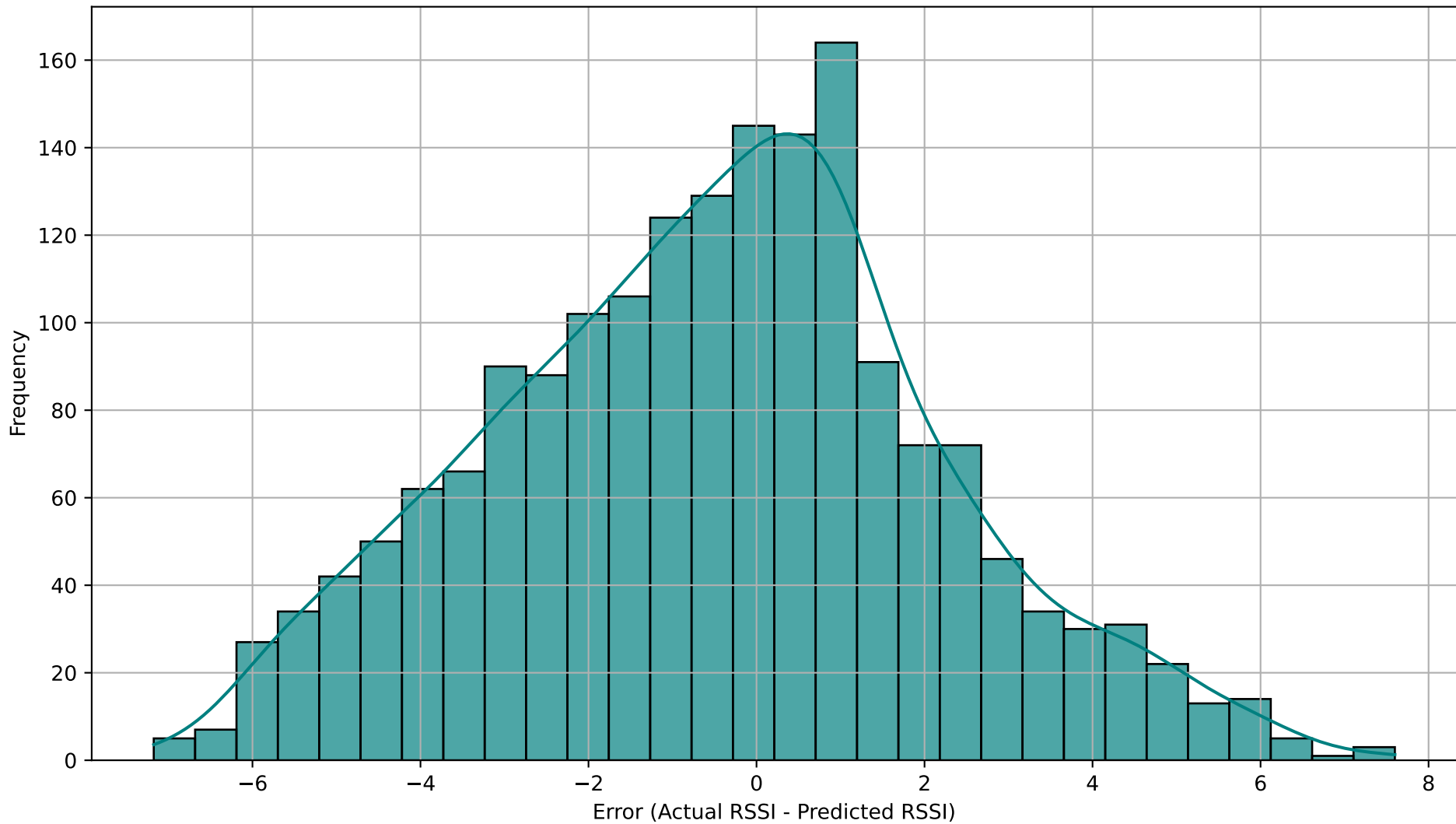
Actual vs Predicted RSSI (Last Step - step 48  $\approx$  4.0 s)



Distribution of Prediction Errors (First Step)

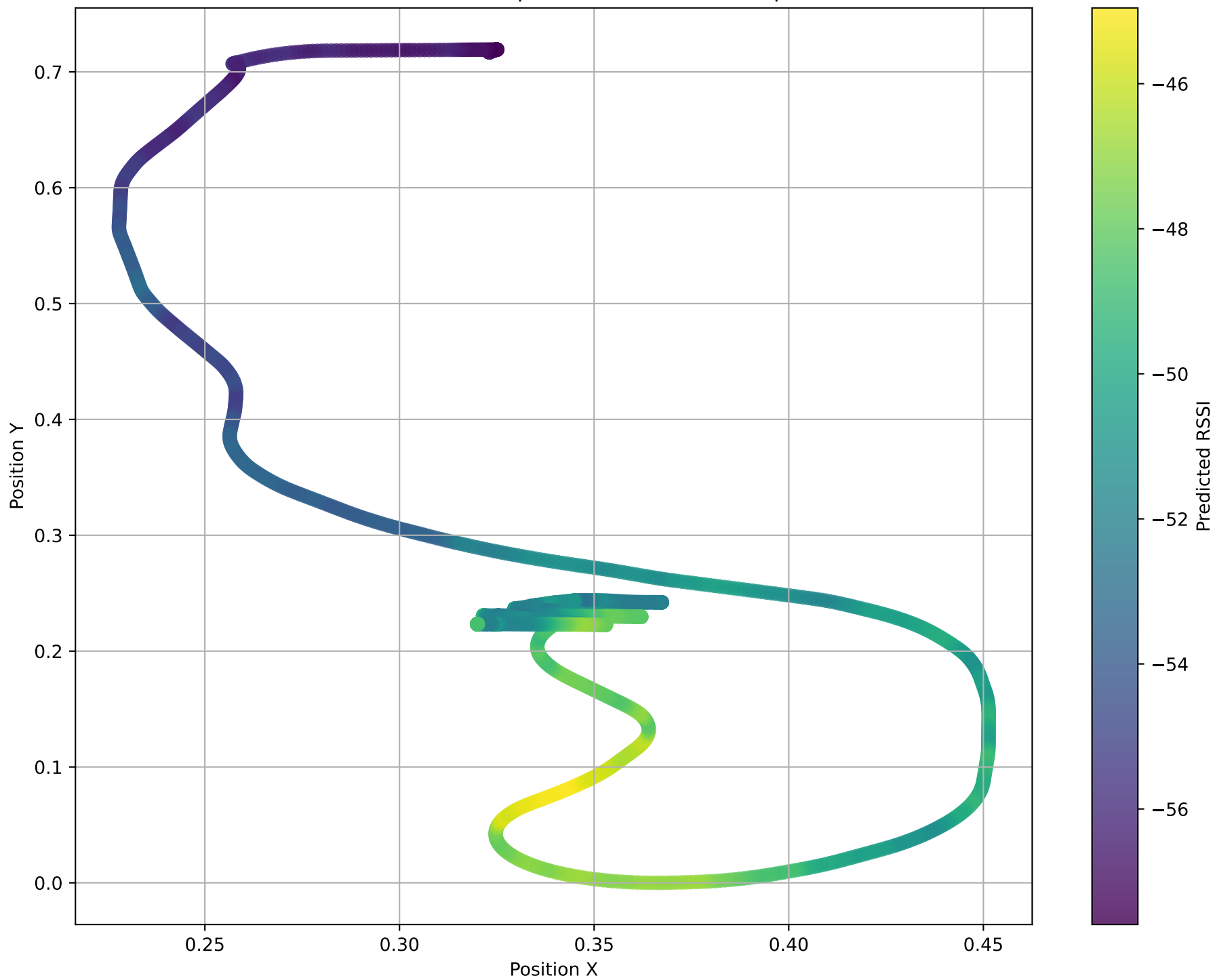


Distribution of Prediction Errors (Last Step - step 48  $\approx$  4.0 s)

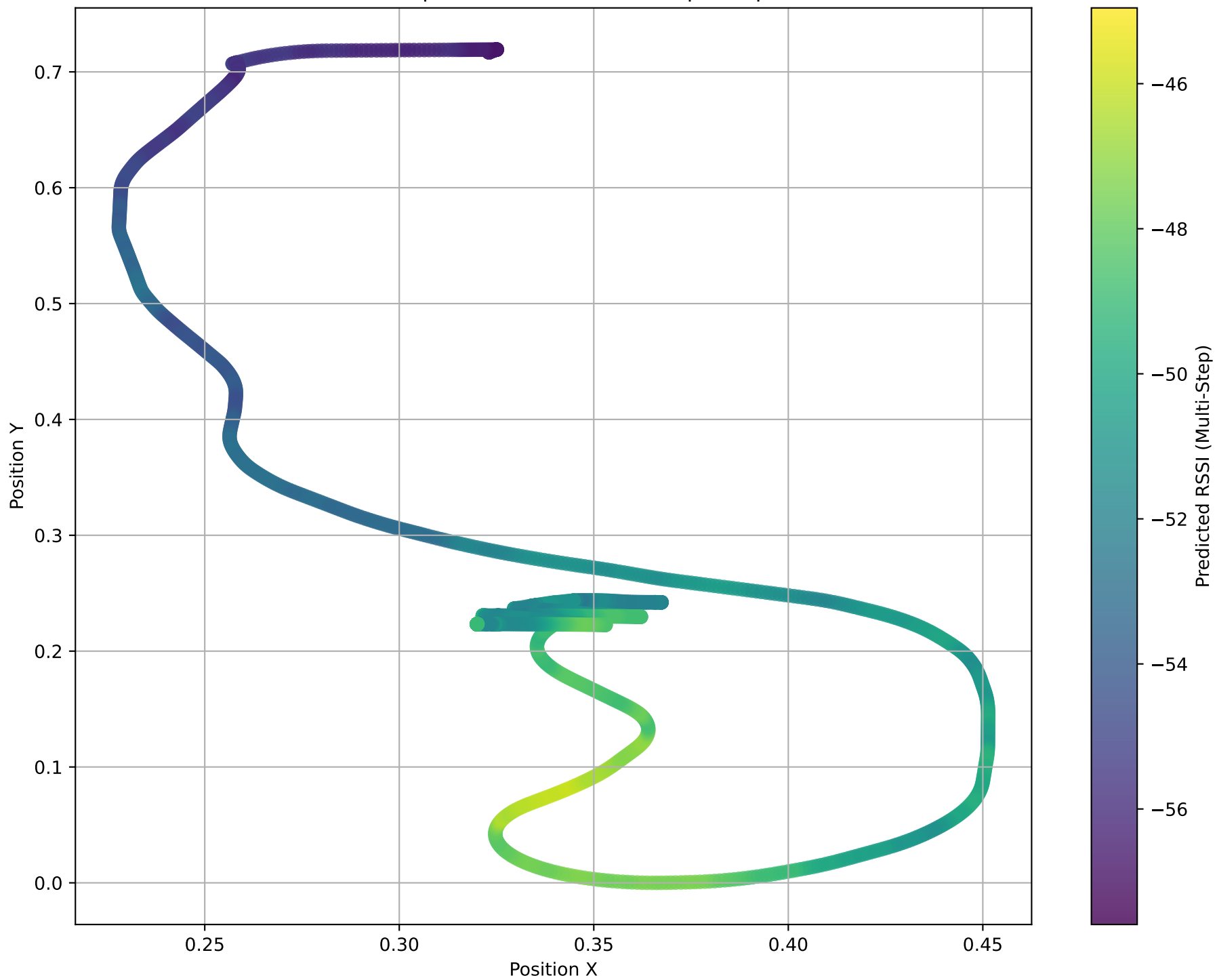




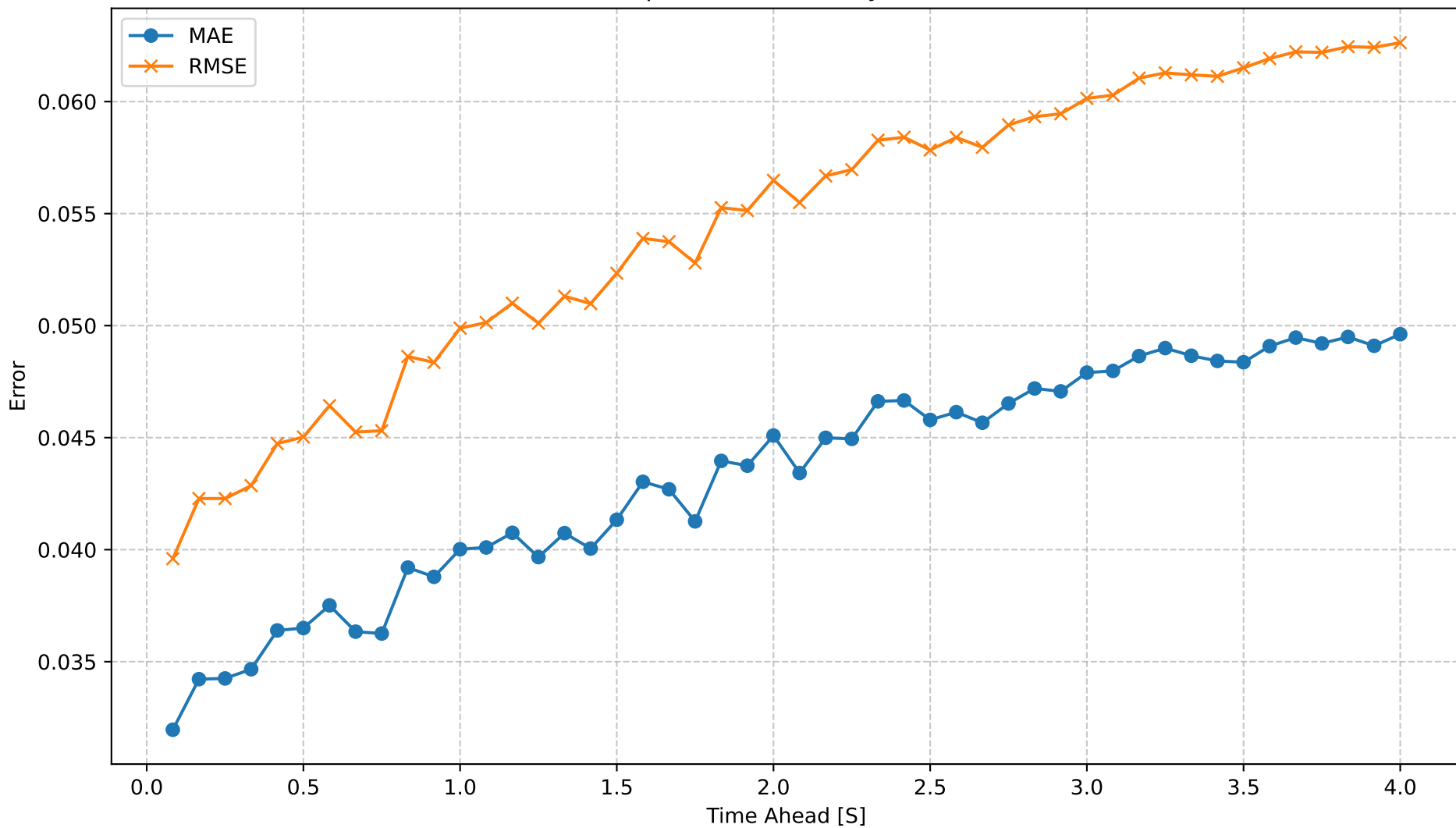
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 48  $\approx$  4.0 s)



Multi-Step Forecast Error by Time Ahead



### Resumo Automático

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Dataset: rssi\_odom\_dataset\_12apr25\_ajust.parquet  
Modelo: LSTM  
Forecast steps: 48

#### MAE / RMSE (dB)

- 1º passo : 0.03 / 0.04
- Mediana : 0.04 / 0.06
- 48º passo : 0.05 / 0.06

Crescimento do erro: 55.2% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 48 (0.05), 46 (0.05), 44 (0.05)

#### Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.